

Harnessing Unstable Underground Wind for Passive LoRa-Based Human Activity Detection in Coal Mines

Zhongxu Bao
China University of Mining and
Technology
Xuzhou, China
Baozx@cumt.edu.cn

Yuqing Yin*
China University of Mining and
Technology
Xuzhou, China
yinyuqing@cumt.edu.cn

Ziyi Xia
China University of Mining and
Technology
Xuzhou, China
xiaziyi@cumt.edu.cn

Xu Yang*
China University of Mining and
Technology
Xuzhou, China
yang_xu@cumt.edu.cn

Chenxi Liu
China University of Mining and
Technology
Xuzhou, China
l_cx@cumt.edu.cn

Qiang Niu
China University of Mining and
Technology
Xuzhou, China
niuq@cumt.edu.cn

Yiwei Zhao
Beijing Jiaotong University
Beijing, China
22221308@bjtu.edu.cn

Abstract

Human activity detection in underground coal mines is challenged by harsh environments and limited energy availability. This study presents a passive sensing system that harnesses unstable underground wind, induced by mine ventilation, to power LoRa-based communication nodes without relying on external power supplies. To mitigate signal intermittency caused by fluctuating wind energy, a Mean Intermittent Amplitude Ratio (MIAR) algorithm is proposed for extracting robust activity features under high interference. The system incorporates adaptive filtering and sliding window techniques to isolate valid signal segments, enabling reliable personnel detection and gesture recognition. Experimental results show that the Random Forest model achieves robust detection performance, while the ResNet model accurately classifies four representative gestures. The solution demonstrates high deployment flexibility, strong resilience to interference, and full energy autonomy, offering a viable approach for non-contact, battery-free sensing in blind zones and disaster-prone areas of underground coal mines.

CCS Concepts

• **Hardware** → **Wireless devices**.

*Both authors are corresponding authors.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp Companion '25, Espoo, Finland

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3754425>

Keywords

LoRa sensing, wind-powered, underground mines, passive detection, MIAR

ACM Reference Format:

Zhongxu Bao, Ziyi Xia, Chenxi Liu, Yuqing Yin, Xu Yang, Qiang Niu, and Yiwei Zhao. 2025. Harnessing Unstable Underground Wind for Passive LoRa-Based Human Activity Detection in Coal Mines. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3754425>

1 Introduction

The underground environment of coal mines is enclosed, complex, and marked by persistent high temperature, humidity, heavy dust, and poor ventilation, all of which pose significant risks to worker safety [12]. In such extreme conditions, establishing a reliable and stable personnel sensing system is critical for ensuring safe production and enhancing emergency response efficiency [16]. However, most existing systems depend on batteries or external power [11], which are prone to failure during emergencies like gas explosions or roof collapses. Once power is lost, monitoring stops and data transmission is interrupted, severely limiting response capabilities.

To address this, we propose a wind-driven LoRa wireless sensing system for continuous personnel monitoring without external power. Underground coal mines possess strong and continuous ventilation airflow [23], which is harnessed by micro wind turbines for autonomous energy harvesting. This energy powers LoRa nodes for data acquisition and transmission, which provides high deployment flexibility and strong adaptability to harsh and power-limited environments.

Although the system is easy to deploy, it faces key technical challenges. Wind speed fluctuations lead to unstable power, and the LoRa transmitter activates only when current exceeds a threshold, causing intermittent signals and missing data. This reduces power

consumption but breaks signal continuity, rendering traditional sampling methods ineffective.

To address this challenge, a Multi-Indicator Amplitude Ratio (MIAR) algorithm is proposed to enhance the quality of the intermittent signals. By amplifying key amplitude features and mitigating variability, MIAR improves the discriminability of signal characteristics, thereby facilitating more robust feature extraction and classification under unstable conditions. Compared to traditional approaches, the MIAR algorithm exhibits greater adaptability and resilience, significantly enhancing personnel detection accuracy. Building on this foundation, the system integrates MIAR-enhanced models for both personnel presence detection and gesture recognition, successfully identifying actions such as forward/backward pushing, waving, and two-handed gestures. These capabilities enable non-contact intention recognition and support human-machine interaction in underground environments.

2 System Design

As shown in Fig. 1, the system first collects signals using a wind-driven LoRa module. The MIAR algorithm enhances signal robustness, followed by signal segment extraction to isolate valid data. These segments are then used for human activity recognition.

2.1 Design of a Wind-Driven LoRa Data Acquisition System

This study designs a wind-driven LoRa data acquisition system to enable continuous sensing in power-free underground environments. The system comprises three modules: wind power generation, signal transmission, and signal reception, enabling autonomous operation without external power.

The wind power module provides the foundation for system operation. It employs a fan-type wind generator that utilizes the continuous ventilation airflow in mines. Equipped with a high-response, low-startup micro turbine structure, the module is capable of generating power even in low-speed and turbulent underground conditions. Operating without batteries or external power, it enables an on-site, wind-triggered self-powered mode. In addition to energy supply, it also triggers system operation. When the generated current reaches the threshold of the communication module, a signal is automatically transmitted without timers or control circuits, demonstrating an “energy-driven behavior.” The system’s operating rhythm is naturally determined by the environmental energy state, offering strong adaptability and responsiveness.

The signal transmission module is directly powered by the wind module and uses LoRa technology for long-range, low-power data communication[14][2][19]. Its triggering is fully driven by the current state, requiring no manual control or scheduled operation, which simplifies the system and improves efficiency.

The signal reception module is deployed in the monitoring area and continuously listens for wireless signals via multi-channel synchronous acquisition. It features high sensitivity and time resolution, allowing extraction of key characteristics such as amplitude and phase from intermittent signals, providing a data foundation for subsequent personnel status recognition and behavior analysis.

The system is compact, easy to deploy, low-power, robust, and highly adaptable to harsh environments. Despite its advantages, the

system faces challenges in practical deployment. Underground wind speed is highly variable, which results in unstable power output, leading to intermittent and unpredictable LoRa signal transmission, increasing processing complexity. The following work will focus on developing signal-level algorithms to address these issues, aiming to enhance system stability, reduce data loss, and improve practical reliability in dynamic underground environments.

2.2 MIAR Algorithm

In the wind-driven LoRa passive sensing system, the signal is intermittent and non-continuous. Traditional feature extraction methods, such as the APR algorithm and conjugate multiplication, have poor stability and are easily affected by interference. To improve robustness, this paper proposes the MIAR algorithm, with its principle and implementation described as follows.

(1) Principle of the MIAR Algorithm

The core idea of the MIAR algorithm is to perform sliding integration on dual-antenna signals before calculating the amplitude ratio to extract stable features.

Assume the dual-antenna received signals are:

$$R_1(t) = |R_1(t)|e^{j\theta_1(t)} \quad (1)$$

$$R_2(t) = |R_2(t)|e^{j\theta_2(t)} \quad (2)$$

where $R_1(t)$ and $R_2(t)$ represent the complex-form signals received by antenna 1 and antenna 2 at time t ; $|R_1(t)|$ and $|R_2(t)|$ are the amplitudes of the received signals at time t ; $\theta_1(t)$ and $\theta_2(t)$ are the instantaneous phases of the signals received by the two antennas; and j is the imaginary unit.

Step 1: Sliding Integration

Given an integration window length T , a small value ϵ is added to avoid division by zero. The window length T should be selected by comprehensively considering noise suppression and signal real-time requirements, usually determined according to actual needs.

The integrated amplitudes are expressed as:

Antenna 1 integrated amplitude:

$$I_{R1}(t) = \sum_{i=t}^{t+T-1} |R_1(i)| + \epsilon \quad (3)$$

Antenna 2 integrated amplitude:

$$I_{R2}(t) = \sum_{i=t}^{t+T-1} |R_2(i)| + \epsilon \quad (4)$$

where $I_{R1}(t)$ and $I_{R2}(t)$ are the sliding integrated amplitudes of the two antennas at time t ; i is the index variable in the sliding integration process.

Step 2: Calculate MIAR Amplitude Ratio Feature

The amplitude ratio feature after sliding integration (denoted as $Amp_{MIAR}(t)$) is finally defined as:

$$Amp_{MIAR}(t) = \frac{I_{R1}(t)}{I_{R2}(t)} = \frac{\sum_{i=t}^{t+T-1} |R_1(i)| + \epsilon}{\sum_{i=t}^{t+T-1} |R_2(i)| + \epsilon} \quad (5)$$

This feature has the following characteristics:

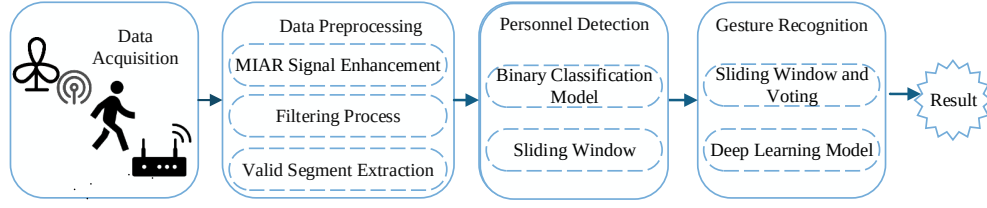


Figure 1: System architecture.

- **Smoothing abrupt changes:** The integration operation is equivalent to low-pass filtering, which can suppress short-term spikes and random interference.
- **Strong anti-discreteness:** In the presence of segment loss, skipped sampling, or intermittent signals, as long as valid data points exist within the window, integration can still be performed to extract effective features.
- **Sensitivity to motion disturbances:** Since disturbances caused by human activity are usually continuous and cumulative in amplitude, the integration value can better capture such trends.

(2) Comparison with the APR Algorithm

The traditional APR[22] algorithm directly uses the amplitude and phase information of the two antennas at the same time to perform calculation.

Amplitude ratio feature:

$$Amp_{APR}(t) = \frac{|R_1(t)|}{|R_2(t)|} \quad (6)$$

Phase difference feature:

$$Phase(t) = \theta_1(t) - \theta_2(t) \quad (7)$$

where $Amp_{APR}(t)$ is the amplitude ratio feature extracted by the APR method, and $Phase(t)$ is the phase difference feature extracted by the APR method.

As shown in Figure 2, the APR method strongly depends on the temporal continuity of the signal and is suitable for continuous signals with stable amplitude and complete sampling. However, in wind-driven scenarios, due to discontinuous signal transmission and severe amplitude fluctuations, APR is prone to feature jitter during real-time calculation, and recognition accuracy drops significantly. In contrast, MIAR performs weighted smoothing through a sliding window and can still extract continuous and stable amplitude ratio features even when signals are interrupted or mutated, significantly enhancing the robustness and practicality of the model.

(3) Comparison with the Conjugate Multiplication Method

Conjugate multiplication[18] is commonly used to extract phase perturbations from complex CSI signals, in the form of:

$$\Delta\theta(t) = \arg(R(t) \cdot R^*(t+1)) \quad (8)$$

where $R(t)$ is the complex signal received at time t ; $R^*(t+1)$ is the complex conjugate of the signal at time $t+1$; $\arg(\cdot)$ denotes the argument (phase angle) of a complex number.

As shown in Figure 2, this method is applicable for estimating phase variations between adjacent time points and is commonly used in WiFi scenarios such as respiration detection and speed analysis. However, it requires a continuous temporal structure of the

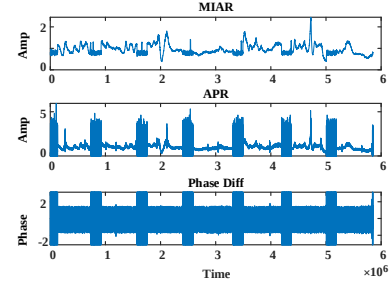


Figure 2: LoRa signal feature comparison.

signal. Once faced with the intermittent transmission of LoRa signals (i.e., gaps between two samples), the method fails or produces random errors. In addition, it is sensitive to hardware parameters such as antenna order and gain, making it unsuitable for deployment in environments with high anti-interference requirements. In contrast, MIAR does not rely on continuity between adjacent time points. Its integration operation can cover multiple valid sampling points and still obtain reliable features even in the presence of intermittency or skipped sampling. Compared with it, MIAR shows better adaptability to the more discrete wind-driven LoRa signals.

2.3 Signal Segment Extraction

To improve the quality of LoRa signal inputs for model training, the system first applies a filtering process, where users can select from four available methods: Kalman filter[21], median filter[7], mean filter[17], and low-pass filter[24]. Each filter type is designed to address specific noise characteristics. The Kalman filter offers adaptive noise reduction in dynamic environments, the median filter is effective at removing impulse noise, the mean filter smooths out minor signal fluctuations, and the low-pass filter suppresses high-frequency components. This flexible preprocessing stage helps reduce unwanted noise while preserving signal structure, laying the foundation for accurate downstream analysis.

Following the filtering stage, the system proceeds to valid segment extraction to isolate structurally meaningful and signal-containing fragments. This step improves data processing efficiency and prevents blank segments, static regions, or high-noise fragments from interfering with recognition results. In implementation, a combination of gradient analysis and sliding-window smoothing [4] is used to identify disturbed regions in the signal. Gradient computation captures the rate of change in signal amplitude and serves to locate potential candidate segments.

Since abrupt changes in gradient values often occur at the boundaries of valid segments, they act as effective indicators for segmentation. However, the raw gradient sequence can contain significant short-term noise. To address this, the system introduces a sliding-window smoothing step, in which a fixed-length window is applied at each observation point. The absolute values of gradients within the window are averaged, effectively suppressing instantaneous disturbances and highlighting more stable fluctuations.

The resulting valid segments—characterized by moderate and continuous variations—are retained as candidate inputs for tasks such as personnel recognition and gesture classification. Given the intermittent and noise-prone nature of LoRa signals, this two-stage process ensures that only informative fragments are preserved, thereby significantly improving the training effectiveness and robustness of the model.

2.4 Human Activity Recognition

Human presence introduces significant disturbances to the signal pattern. Leveraging this characteristic, we trained a binary classification model to distinguish between presence and absence of personnel. During inference, we segment the incoming signal using a sliding window and classify each segment independently to achieve real-time, localized detection under intermittent signal conditions.

For gesture recognition, we construct a labeled dataset and then train several models to identify representative gestures. However, due to the complexity of gesture signals, non-gesture segments may sometimes resemble valid gestures in feature space, leading to ambiguous predictions. To resolve this, we introduced a voting mechanism [3]. After classifying all segments, we aggregate the results and select the gesture label with the highest vote count as the final output. This strategy improves recognition stability and mitigates the effects of noise and transitional segments, enhancing robustness in dynamic environments.

3 Experiment Evaluation

3.1 Experiment Setup

The experimental setup, shown in Figure 3, was deployed in an underground coal mine laboratory to evaluate the proposed passive LoRa system. A fan-type micro wind turbine simulated mine ventilation and powered an SX1276 LoRa transmitter for self-sustained signal transmission. A USRP X310 receiver captured the intermittent LoRa signals with high sensitivity for further analysis under realistic underground conditions.

3.2 Human Activity Recognition Results

In the personnel detection experiment, six classification models were built based on a manually labeled dataset: Random Forest (RF)[13], Multi-Layer Perceptron (MLP)[6], Naive Bayes (NB)[1], K-Nearest Neighbors (KNN)[15], Support Vector Machine (SVM)[9], and Logistic Regression (LR)[10]. All models were trained and evaluated using five-fold cross-validation. Figure 4 illustrates that MLP achieved an accuracy of 96.69%, NB 87.87%, KNN 97.06%, SVM 94.49%, and LR 95.59%. RF outperformed all others with an accuracy of 98.53%, and was therefore selected as the deployed model in the system.

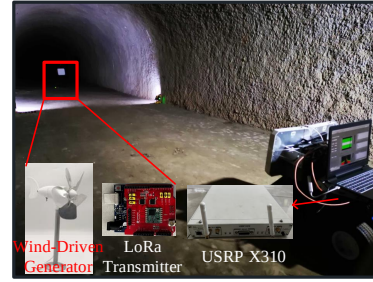


Figure 3: Wind-Driven LoRa data acquisition setup.

To evaluate model adaptability in complex scenarios, three types of personnel status experiments were conducted: varying wind speed conditions (low, medium, high), different movement states (slow walking, normal walking, fast running), and different numbers and body types of individuals (single, double). Results showed that the RF model maintained over 97% accuracy across all test conditions, demonstrating strong stability and generalization ability.

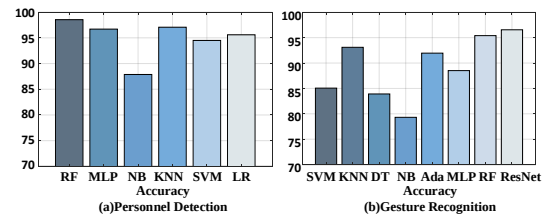


Figure 4: Experiments results.

For gesture recognition, four typical actions were selected: forward/backward push, left/right wave, up/down wave, and double-hand push. Eight models were evaluated: SVM, KNN, Decision Tree[5], NB, Adaboost[8], MLP, RF, and ResNet[20]. Results showed that NB performed the worst (accuracy 79.31%), while ResNet achieved the best performance with 96.55% accuracy, and was selected as the core model for the gesture recognition module.

In summary, the RF-based personnel detection model and the ResNet-based gesture recognition model both demonstrated excellent accuracy and adaptability under intermittent LoRa signal conditions, providing reliable technical support for passive intelligent sensing in complex underground coal mine environments.

4 Conclusion

This work proposes a wind-powered LoRa sensing system for underground mine rescue scenarios with limited power supply. A wind-driven generator enables signal acquisition, while the MIAR method enhances intermittent signal features. A robust processing pipeline supports non-contact personnel detection and gesture recognition, offering a sustainable and efficient solution for sensing in complex environments.

Acknowledgments

This study was supported by Natural Science Foundation of Jiangsu Province (BK20221109 and BK20231059)

References

- [1] Daniel Berrar. 2025. Bayes' theorem and naive Bayes classifier. (2025).
- [2] Zhaoxin Chang, Fusang Zhang, Jie Xiong, Junqi Ma, Beihong Jin, and Daqing Zhang. 2022. Sensor-free soil moisture sensing using lora signals. *Proceedings of the 2022 7th International Conference on Machine Learning Technologies* 6, 2 (2022), 1–27.
- [3] Mejdil Dallel, Vincent Havard, Yohan Dupuis, and David Baudry. 2022. A sliding window based approach with majority voting for online human action recognition using spatial temporal graph convolutional neural networks. In *Proceedings of the 2022 7th International Conference on Machine Learning Technologies*. 155–163.
- [4] Anne M Denton. 2023. Multiscale Aggregation Over Sliding Windows. In *Spatial Gems, Volume 2*. 29–36.
- [5] Riccardo Guidotti, Anna Monreale, Mattia Setzu, and Giulia Volpi. 2024. Generative model for decision trees. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 21116–21124.
- [6] Qiuyu Huang, Zequn Jie, Lin Ma, Li Shen, and Shenqi Lai. 2025. A Pyramid Fusion MLP for Dense Prediction. *IEEE Transactions on Image Processing* (2025).
- [7] Elvin Isufi, Fernando Gama, David I Shuman, and Santiago Segarra. 2024. Graph filters for signal processing and machine learning on graphs. *IEEE Transactions on Signal Processing* 72 (2024), 4745–4781.
- [8] Xue Jiang, Yuan Xu, Wei Ke, Yang Zhang, Qun-Xiong Zhu, and Yan-Lin He. 2022. An imbalanced multifault diagnosis method based on bias weights AdaBoost. *IEEE Transactions on Instrumentation and Measurement* 71 (2022), 1–8.
- [9] Zhihui Lai, Guangfei Liang, Jie Zhou, Heng Kong, and Yuwu Lu. 2024. A joint learning framework for optimal feature extraction and multi-class SVM. *Information Sciences* 671 (2024), 120656.
- [10] Xingyu Lu, Hasin Us Sami, and Başak Güler. 2023. SCALR: Communication-efficient secure multi-party logistic regression. *IEEE Transactions on Communications* 72, 1 (2023), 162–178.
- [11] Soumyadeep Paty, Arindam Biswas, Sonia Djebali, Guillaume Guerard, and Supreeti Kamilya. 2025. IoT-Enabled Methane Monitoring and LSTM-Based Forecasting System for Enhanced Safety in Underground Coal Mining. *ACM Transactions on Internet of Things* 6, 1 (2025), 1–29.
- [12] S Rajan, Neelam Sanjeev Kumar, Gokul Chandrasekaran, K Sowmiya, K Varalakshmi, Neeraj Priyadarshi, MS Bhaskar, and Niraj Kumar. 2022. Atmospheric monitoring and fire prevention electronic system for mining workers. In *2022 IEEE 9th Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering (UPCON)*. IEEE, 1–5.
- [13] Jake S Rhodes, Adele Cutler, and Kevin R Moon. 2023. Geometry-and accuracy-preserving random forest proximities. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 9 (2023), 10947–10959.
- [14] Zehua Sun, Huanqi Yang, Kai Liu, Zhimeng Yin, Zhenjiang Li, and Weitao Xu. 2022. Recent advances in LoRa: A comprehensive survey. *ACM Transactions on Sensor Networks* 18, 4 (2022), 1–44.
- [15] Shahadat Uddin, Ibtisham Haque, Haohui Lu, Mohammad Ali Moni, and Ergun Gide. 2022. Comparative performance analysis of K-nearest neighbour (KNN) algorithm and its different variants for disease prediction. *Scientific Reports* 12, 1 (2022), 6256.
- [16] Chunling Wang and Haoran Chen. 2021. Design of coal mine safety monitoring system based on multi-agent. In *2021 2nd International Conference on Artificial Intelligence and Information Systems*. 1–7.
- [17] Ge Wang, Haofan Cai, Chen Qian, Jinsong Han, Shouqian Shi, Xin Li, Han Ding, Wei Xi, and Jizhong Zhao. 2020. Hu-fu: Replay-resilient rfid authentication. *IEEE/ACM Transactions on Networking* 28, 2 (2020), 547–560.
- [18] Dan Wu, Youwei Zeng, Ruiyang Gao, Shenjie Li, Yang Li, Rahul C Shah, Hong Lu, and Daqing Zhang. 2021. WiTraj: Robust indoor motion tracking with WiFi signals. *IEEE Transactions on Mobile Computing* 22, 5 (2021), 3062–3078.
- [19] Binbin Xie, Minhao Cui, Deepak Ganesan, and Jie Xiong. 2024. Wall matters: Rethinking the effect of wall for wireless sensing. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 4 (2024), 1–22.
- [20] Kexin Yao, Han Li, Ming Liu, Bo Gao, Ke Xiong, and Pingyi Fan. 2023. ARES-WiGR: An Attention-enhanced ResNet based Wi-Fi Gesture Recognition. In *IEEE INFOCOM 2023-IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*. IEEE, 1–2.
- [21] Yuan Yuan, Gang Qin, Da Li, Min Zhong, Yingchun Shen, and Yongzhong Ouyang. 2024. Real-time joint filtering of gravity and gravity gradient data based on improved Kalman filter. *IEEE Transactions on Geoscience and Remote Sensing* (2024).
- [22] Fusang Zhang, Zhaoxin Chang, Jie Xiong, Rong Zheng, Junqi Ma, Kai Niu, Beihong Jin, and Daqing Zhang. 2021. Unlocking the beamforming potential of lora for long-range multi-target respiration sensing. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 5, 2 (2021), 1–25.
- [23] Huili Zhang, Binghao Li, Mahmoud Karimi, Serkan Saydam, and Mahub Hassan. 2023. Recent advancements in IoT implementation for environmental, safety, and production monitoring in underground mines. *IEEE Internet of Things Journal* 10, 16 (2023), 14507–14526.
- [24] Huan Zhao, Yingxue Gao, Haijiao Chen, Bo Li, Guanghui Ye, and Zixing Zhang. 2025. Enhanced Multimodal Emotion Recognition in Conversations via Contextual Filtering and Multi-Frequency Graph Propagation. In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 1–5.